hecthese: the LaTeX document class for dissertations and theses at HEC Montréal

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Abstract

The **hecthese** LaTeX document class has been created to allow graduate students at HEC Montréal to write their dissertation or thesis with the document preparation system while complying to all presentation standards required by the University. As such, the class complies entirely with the *Guidelines for Writing an Academic Work at a Graduate Level*[2], hereafter called the *Guidelines*.

1 Installing the class

1.1 Prerequisites

The use of this document class presumes that you have already installed a TeX distribution and a code editor. For the development of **hecthese**, the TeXLive 2016 distribution[3] and the TeXStudio code editor[4] were used. Its features were tested with the `latex`, `pdflatex`, `bibtex` and `makeindex` compilers. The class has also been tested with the MiKTeX distribution[1] and WinEdt code editor, with the 2017 to 2023 versions of TeXLive as well as with the Overleaf web application. We invite you to test the class with your own distribution and code editor and to use it like you would with any other work you do with LaTeX.

1.2 Installation

> **Overleaf and MacOS users**
> For a reason still unresolved, the installation process described hereunder doesn't work on MacOS. You must **download this preinstalled version** of the class to work with it.
>
> If you are an Overleaf user, please use the preinstalled version mentioned above to start a project.

The `.zip` archive you downloaded contains the following files and directory:

1. **hecthese-fr.ins**: the french class installation file;

[1] A big "thank you!" goes to Franck Jeannot for making the tests.

2. **hecthese-en.ins**: the english class installation file;
3. **hecthese.dtx**: the bilingual documented source code;
4. **hecthese.pdf**: the french version of the class documentation;
5. **hecthese-en.pdf**: the english version of the class documentation;
6. **README.md**: the file required to show the class description on the ctan.org Website.

Follow these steps to install the class [2]:

1. Create a working directory on your computer.
2. Extract the `.zip` archive in that directory.
3. Open a command-line editor.
4. Change directories until you reach your working directory.
5. Type the following command in the editor:

```
latex hecthese-en.ins
```

The command will create a bunch of files that will become your document. The list of files needed will depend on the type of document you'll be writing.

1.3 List of a document's required files

Table 1 shows a list of all required files for each type of document. Depending on the type of work you'll be doing, you'll have to choose between one of these four templates:

- **template-phd-classic.tex** : template for writing a dissertation in a classic manner;
- **template-phd-articles.tex** : template for writing a dissertation with articles;
- **template-msc-classic.tex** : template for writing a thesis in a classic manner;
- **template-msc-articles.tex** : template for writing a thesis with articles.

Once you have chosen a template file, delete the three remaining ones as you won't need them. The template file is your document's master file, the one on which you will have to run all compilers needed to generate your final document.

[2] An installation tutorial is available at https://www.youtube.com/watch?v=nfTEgcJbufs (in French with English subtitles).

Table 1: List of a document's required files

File	Dissertations		Theses	
	Classic	With articles	Classic	With articles
template-phd-*.tex	✔	✔	⊘	⊘
template-msc-*.tex	⊘	⊘	✔	✔
abstract-french.tex	✔	✔	✔	✔
abstract-english.tex	✔	✔	✔	✔
acronym-list.tex	✔	✔	✔	✔
dedication.tex	✔	✔	⊘	⊘
acknowledgements.tex	✔	✔	✔	✔
preface.tex	✔	✔	✔	✔
introduction.tex	✔	✔	✔	✔
theoretical-framework.tex	⊘	✔	⊘	⊘
literature-revue.tex	⊘	⊘	✔	✔
chapter-1.tex	✔	⊘	✔	⊘
chapter-2.tex	✔	⊘	✔	⊘
chapter-3.tex	✔	⊘	✔	⊘
article-1.tex	⊘	✔	⊘	✔
article-2.tex	⊘	✔	⊘	⊘
article-3.tex	⊘	✔	⊘	⊘
conclusion.tex	✔	✔	✔	✔
appendix.tex	✔	✔	✔	✔
student-contribution.tex	⊘	⊘	✔	✔
certificate.tex	⊘	⊘	✔	✔

It is also in this file that you will write all the metadata related to your work (title, author, etc.) and add all packages, user-defined commands and environments you will need to complete your work.

Your work will be written in the other files. Each file represents a section of your work. Their name is, as such, self-explanatory. You can add and delete as many files as you wish. Just make sure to include the added files to your template file with `\include` commands, to erase the `\include` commands related to the deleted files and, most of all, to comply with all the *Guidelines*' rules related to the mandatory sections of your work and their presentation order.

2 Using the class

The **hecthese** document class has been created to be as simple and as flexible as possible. The main goal of the class is to layout your dissertation or thesis according to the presentation standards required by the *Guidelines* and to allow you to add as many features as you see fit.

hecthese is based on the `memoir` document class[5] which already provides numerous features. Before programming any new commands or environments,

you should read the class' exhaustive documentation [3].

2.1 Class options

The class has very few default options. They only concern font size, document language and type. They are all enclosed in the `\documentclass` command.

`10pt,11pt,12pt` The french edition of the *Guidelines* mentions that "font size usually varies between 10 and 12 points in the text"[1, p. 15]. You have to choose between one of these three options: `10pt`, `11pt` or `12pt`. If none of the options is chosen, the default `12pt` option will be used by the class [4].

`english,french` `french` and `english` options come from the `babel` package[6] and they are used to indicate to the compiler in which language(s) the document is written. The last language in the options list is the document's default language. Even if these options come from the `babel` package, they are inserted in the `\documentclass` command so they can be applied globally to all packages that could use them.

`phdclassique` `phdarticles` `mscclassique` `mscarticles` You use `phdclassique` when you write a dissertation in a classic manner, `phdarticles` when you write a dissertation with articles, `mscclassique` when you write a thesis in a classic manner and `mscarticles` when you write a thesis with articles. Each of these options is automatically inserted in its corresponding template file when you install the **hecthese** document class. Since `phdclassique` is the class' default document type option, you could erase this option from the **template-phd-classic.tex** file, but we advise you don't do that so you don't encounter problems while writing.

As for the three other document type options, **never erase them from the `\documentclass` command!** By doing so, you would encounter errors when compiling your document, especially during the BiBTEX compilations and during the title page layout.

2.2 Class commands

The **hecthese** document class provides commands that can be separated into two categories:

1. document metadata (author, title, etc.) used to generate the title pages;
2. document layout commands.

2.2.1 Document metadata

The following commands are used in all document types.

`\HECtitle` This is the title of your dissertation or thesis. Do not use the LaTeX `\title` command as the class will not take it into account. If your title is very long, separate it in more than one line with the \\ command.

`\HECsubtitle` This is the subtitle of your dissertation or thesis. If there is a subtitle, a colon

[3] . . . although. . . the author of the present class did not read it himself entirely. *Mea culpa*. . .
[4] As an example, this documentation is written with a `10pt` font size

(:) will automatically separate the title from the subtitle, and the latter will be laid out below the title without you having to add the \\ command after the title.

`\HECauthor` This is the author of the dissertation or thesis, meaning you, unless you have plagiarized your work... Write your name in the *FirstName LastName* format. Do not use the LaTeX `\author` command as the class will not take it into account.

`\HECoption` The `\HECoption` determines your dissertation's or thesis' option.

`\HECsubMonth` `\HECsubYear` `\HECsubMonth` and `\HECsubYear` represent the month and year of the final submission of your work. Write down the month in full letters and the year in the *YYYY* format.

`\HECpdfauteur` `\HECpdftitre` These are variants of the `\HECauthor` and `\HECtitle` commands and they are used for inserting metadata in the final .pdf document. They are included in the `\hypersetup` command's `pdftitle` and `pdfauthor` options provided by the `hyperref` package[7].

The following commands are used in dissertations only. If you are writing a thesis, you can skip to Section 2.2.2.

`\HECrapporteurPresident` `\HECresearchDirector` `\HECresearchCodirector` `\HECexternalexaminator` `\HECdirectorRepresentative` Each of these commands identifies a person involved in your work. The commands take two arguments. The first argument is the person's name in the *FirstName LastName* format. The second argument is the person's gender, male (M) or female (F). It is important to type in the person's gender as it will affect how the person's title is rendered on the title page, and it is not advisable to hurt your directors' feelings upon submission of your work...

`\HECjuryMember` This command has the same role as the five preceding ones, but it only takes one argument, the person's name, as the job title is in this case gender-neutral.

`\HECcodirectorUniversity` `\HECjuryMemberUniversity` `\HECexaminatorUniversity` By default, your codirector, jury member and external examinator are considered working at HEC Montréal by the **hecthese** document class, but they could be working at another university. You'll then have to modify the university's name is these commands when necessary.

2.2.2 Document layout

`\HECtitlepages` Title pages differ from one document type to the other as it is shown in the *Guidelines*' title page templates. A jury identification page is also included after the title page in dissertations. The `\HECtitlepages` command automatically generates all title pages using the contents of the commands listes in Section 2.2.1.

> **Why are my title pages displayed in French?**
> The Université de Montréal's *Règlement pédagogique de la faculté des études supérieures et postdoctorales*[15, art. 88,89,134,135] states the language and presentation standards that prevail for dissertations and thesis, and refers to the *Guide de présentation des mémoires et des thèses*[16, p. 22] where it is written that "*title pages must be written in French even when the dissertation or thesis is written in English*". Since HEC Montréal is one of Université de Montréal's affiliate universities, this rule prevails.

`\HECtitreIntroduction` `\HECtitreConclusion` `\HECgenererTitres` The `\HECtitreIntroduction` and `\HECtitreConclusion` commands are placed as arguments in the **introduction.tex** and **conclusion.tex** files' `\chapter*`

command. They are used in the title generation with the `\HECgenererTitres` command. If you write a dissertation or thesis in a classical manner, the titles of your document's introduction and conclusion will be "Introduction" and "Conclusion", whereas if you write a dissertation or thesis with articles, the titles will be "General Introduction" and "General Conclusion".

`\HECtdmAbreviations`
`\HECtdmRemerciements`
`\HECtdmAvantPropos`
`\HECtdmCadreTheorique`
`\HECtdmRevueLitterature`
`\HECtdmResumeArticle`

Many parts of HEC Montréal's dissertations and theses being "homemade" sections and chapters, no official english translations exist for their titles. The **hecthese** class takes care of the problem by automatically detecting the document's default language and by generating the appropriate titles for these sections: general introduction and conclusion, acronym list, acknowledgements, preface, theoretical framework, literature revue and each article's abstract.

2.3 Class environments

The **hecthese** class' environments all share the same purpose : laying out your work.

`HECdedication` (*env.*)

This environment can be found in the **dedication.tex** file. It is inside the `HECdedication` environment that you will write... your dedication. At the document's compilation, the dedication will be vertically centered, aligned at the right margin and emphasized.

`HECabbreviations` (*env.*)

`HECabreviations` is a variant of the `description` environment. It is used to build your acronyms list. It takes as argument the longest of your acronyms and uses its length to align the list in two columns. The `calc`[8] and `enumitem`[9] packages are used to layout the list.

3 Writing your dissertation or thesis

In this section, we will give you a few hints for writing your dissertation or thesis with the **hecthese** document class.

3.1 One document divided in multiple files

As mentioned in Section 1.3, your document is divided in many files. A few reasons underlie the choice of such a division:

1. Separate your code (packages, commands, environments, etc.) from your document;
2. Reduce the files' size;
3. Facilitate searching through your text;
4. Offer better flexibility for adding or deleting sections.

Each file is heavily commented with instructions to make sure you don't make any mistake or mess with the document structure. Carefully read through the comments and delete them if necessary.

3.2 Bibliographies and citations

The document class uses the `biblatex` package[17] to manage the dissertations and theses' citations and bibliographies.

Previous versions of `hecthese` used the `chapterbib` package, but it was incompatible with the `memoir` document class.

The `biblatex` package and `biber` compiler being more versatile and flexible that their `bibtex` counterpart, they have become the default bibliography managers starting at `hecthese` version 2.0.

Please read the Getting started with BibLaTeX guide to learn how to use the package and look at the class' source code to see how it has been implemented.

What is important to understand is that in dissertations and theses written with articles, each article has to have its own bibliography and the entire document has to have a global bibliography regrouping references found everywhere, including in the articles.

To do so, you must add `keywords` in your `.bib` file's references, like `article-n` or `general-bib` for example, and write the appropriate `keyword` in the `\printbibliography[]` commands found in the `template` and `articles` files.

4 Compilation

> **Overleaf users**
> Since Overleaf only has a single "Compile" button, this section doesn't concern you.

When the time comes for compiling your document, simply pressing "Compile" in your code editor won't suffice. A precise chain of compilations is necessary if you want your document to be rendered properly [5].

Here is the compilation chain you need to do in order to generate your classic dissertation or thesis. You can perform these compilations from your code editor or from the command line. In the following list, replace `*` by `msc` or `phd` and `**` by `classic` ou `articles` depending of the type of document you're writing.

- `pdflatex template-*-**.tex`
- `biber template-*-**.tex`
- `makeindex template-*-**.tex` [6]
- `pdflatex template-*-**.tex`
- `pdflatex template-*-**.tex`

[5] A video tutorial showing the compilation process is available at https://www.youtube.com/watch?v=hS3LMC3H55w (in French with English subtitles)

[6] Do this compilation only if you,ve inserted index entries and the `\printindex` command in your document.

Compiling a dissertation or thesis is pretty straightforward. All compilation steps are performed on your template file. You run a first compilation with `pdflatex`, you generate your bibliography and index with `biber` and `makeindex` and you recompile your template file at least twice with `pdflatex` in order to output the bibliography, the index and the table of contents.

When compiling your document using the command line, you don't need to type the file extension (`tex`). Only the file name will suffice.

A hecthese class source code

You will find in this section the LaTeX **hecthese** document class' source code. If you want to see how it is programmed, help debug or enhance it, etc., this section is for you.

A.1 Boolean tests and values

To run conditional tests, the class uses the `ifthen` package[10]. Boolean variables are used to determine what kind of document is being written (Ph.D. or M.Sc., classic or by articles) and to determine each contributor's gender. Once the variables have been instantiated, default values are given to these variables.

```
1 ⟨*class⟩
2 \RequirePackage{ifthen}
3
4 % Booleens
5 \newboolean{HEC@isPhD}      % Le travail est une these ou non
6 \newboolean{HEC@isClassique}    % Le travail est redige de maniere classique ou non
7 \newboolean{HEC@isPresRappFemme}   % President rapporteur femme ou non
8 \newboolean{HEC@isDirRechFemme}    % Directeur de la recherche femme ou non
9 \newboolean{HEC@isCodirRechFemme}   % Codirecteur de la recherche femme ou non
10 \newboolean{HEC@isExamExtFemme}    % Examinateur externe femme ou non
11 \newboolean{HEC@isRepDirFemme}    % Representant du directeur femme ou non
12
13 % Valeurs par defaut
14 \setboolean{HEC@isPhD}{true}
15 \setboolean{HEC@isClassique}{true}
16 \setboolean{HEC@isPresRappFemme}{false}
17 \setboolean{HEC@isDirRechFemme}{false}
18 \setboolean{HEC@isCodirRechFemme}{false}
19 \setboolean{HEC@isExamExtFemme}{false}
20 \setboolean{HEC@isRepDirFemme}{false}
```

A.2 Class options

The few options available with the class are declared hereunder. Note that the `babel` package's options are not declared here.

```
21
22 % Taille de la police de caractere
```

```
\DeclareOption{10pt}{%
 \PassOptionsToClass{10pt}{memoir}
}
\DeclareOption{11pt}{%
 \PassOptionsToClass{11pt}{memoir}
}
\DeclareOption{12pt}{%
 \PassOptionsToClass{12pt}{memoir}
}

% Type de document
\DeclareOption{mscclassique}{%
 \setboolean{HEC@isPhD}{false}
 \setboolean{HEC@isClassique}{true}
}
\DeclareOption{mscarticles}{%
 \setboolean{HEC@isPhD}{false}
 \setboolean{HEC@isClassique}{false}
}
\DeclareOption{phdclassique}{%
 \setboolean{HEC@isPhD}{true}
 \setboolean{HEC@isClassique}{true}
}
\DeclareOption{phdarticles}{%
 \setboolean{HEC@isPhD}{true}
 \setboolean{HEC@isClassique}{false}
}

```

A.3 Class loading

The class is loaded in the document with all the options declared by the end user. If a font size is not specified, the classe loads the `12pt` font size by default.

```

% Chargement de la classe
\DeclareOption*{\PassOptionsToClass{\CurrentOption}{memoir}}
\ExecuteOptions{12pt}      % Taille par defaut
\ProcessOptions
\LoadClass{memoir}

```

A.4 Required packages

Very few packages are loaded in the class so you can write your document with a maximum flexibility.

The class uses the `natbib` package[12] so you can write your citations in the *author-year* format. The `chapterbib` package is loaded only if the dissertation or thesis is written with articles.

The other packages loaded are typical to any other document: file encoding,

graphics, images and color management, math modes, etc.

```
58
59 \RequirePackage[utf8]{inputenc}    % Pour ecrire les diacritiques directement
60 \RequirePackage[T1]{fontenc}    % Utilisation des polices T1
61 \RequirePackage{babel}      % Support multilingue
62 \RequirePackage{csquotes}     % Requis par babel
63 \RequirePackage[autolanguage]{numprint}
64 \RequirePackage{calc}      % Necessaire pour la liste des abreviations
65 \RequirePackage{enumitem}     % Necessaire pour la liste des abreviations
66 \RequirePackage{tocvsec2}     % Pour determiner la profondeur de la TDM
67 \RequirePackage{graphicx}     % Insertion de graphiques et d'images
68 \RequirePackage{color}      % Gestion des couleurs
69 \RequirePackage{amsmath}     % Package obligatoire pour les maths
70 \RequirePackage{iflang}      % Detection de la langue
71
```

A.5 Layout

All the presentation standards required by the *Guidelines* are programmed hereunder. During compilation, LaTeX will complain that some headers are too small for your content, but this will not generate errors (the compiler only issues warnings).

```
72
73 \pagestyle{plain}       % Numero de page centre au pied de page
74 \renewcommand{\baselinestretch}{1.5}  % Interligne et demie
75 \setlength{\topmargin}{0cm}     % Marge du haut
76 \setlength{\oddsidemargin}{1.5cm}   % Marge de gauche des pages impaires
77 \setlength{\evensidemargin}{1.5cm}   % Marge de gauche des pages paires
78 \setlength{\textwidth}{15cm}    % Largeur du bloc de texte
79 \setlength{\textheight}{21.9cm}    % Hauteur du bloc de texte
80 \setlength{\marginparwidth}{0pt}   % Suppression des notes de marge
81 \setlength{\marginparsep}{0pt}    % Suppression du separateur de marge
82 \setlength{\headheight}{0pt}    % Suppression de l'entete
83 \setlength{\headsep}{0pt}     % Suppression du separateur d'entete
84
```

A.6 Class commands

A.6.1 Document metadata

Every metadata-related command that you'll find in the document preamble has its internal command equivalent. Example given, the `\HECtitre` has `\HEC@titre` as equivalent. The internal commands are used to generate the document's title page and jury indentification page.

Many commands have their english translation so english-speaking students can work easily with the class, but these english translations are only placeholders that call their french equivalent.

```
85
86 % Commandes internes
87 \newcommand{\HEC@titre}{}
```

```
\newcommand{\HEC@sousTitre}{}
\newcommand{\HEC@auteur}{}
\newcommand{\HEC@optionPhD}{}
\newcommand{\HEC@optionMSc}{}
\newcommand{\HEC@moisDepot}{}
\newcommand{\HEC@anneeDepot}{}
\newcommand{\HEC@presidentRapporteur}{}
\newcommand{\HEC@directeurRecherche}{}
\newcommand{\HEC@codirecteurRecherche}{}
\newcommand{\HEC@universiteCodirecteur}{}
\newcommand{\HEC@membreJury}{}
\newcommand{\HEC@universiteMembreJury}{}
\newcommand{\HEC@examinateurExterne}{}
\newcommand{\HEC@universiteExaminateur}{}
\newcommand{\HEC@representantDirecteur}{}

% Commandes publiques
\newcommand{\HECtitre}[1]{%
 \renewcommand{\HEC@titre}{#1}
}
\newcommand{\HECtitle}[1]{\HECtitre{#1}}

\newcommand{\HECsoustitre}[1]{%
 \renewcommand{\HEC@sousTitre}{#1}
}
\newcommand{\HECsubtitle}[1]{\HECsoustitre{#1}}

\newcommand{\HECauteur}[1]{%
 \renewcommand{\HEC@auteur}{#1}
}
\newcommand{\HECauthor}[1]{\HECauteur{#1}}

\newcommand{\HECoption}[1]{%
 \ifthenelse{\boolean{HEC@isPhD}}{%
  \renewcommand{\HEC@optionPhD}{#1}
 }{%
  \renewcommand{\HEC@optionMSc}{#1}
 }
}

\newcommand{\HECmoisDepot}[1]{%
 \renewcommand{\HEC@moisDepot}{#1}
}
\newcommand{\HECsubMonth}[1]{\HECmoisDepot{#1}}

\newcommand{\HECanneeDepot}[1]{%
 \renewcommand{\HEC@anneeDepot}{#1}
}
\newcommand{\HECsubYear}[1]{\HECanneeDepot{#1}}

\newcommand{\HECpresidentRapporteur}[2]{%
 \renewcommand{\HEC@presidentRapporteur}{#1}
 \ifthenelse{\equal{#2}{F}}{%
  \setboolean{HEC@isPresRappFemme}{true}
```

```
 }{%
  \setboolean{HEC@isPresRappFemme}{false}
 }
}

\newcommand{\HECrapporteurPresident}[2]{\HECpresidentRapporteur{#1}{#2}}
\newcommand{\HECdirecteurRecherche}[2]{%
 \renewcommand{\HEC@directeurRecherche}{#1}
 \ifthenelse{\equal{#2}{F}}{%
  \setboolean{HEC@isDirRechFemme}{true}
 }{%
  \setboolean{HEC@isDirRechFemme}{false}
 }
}
\newcommand{\HECresearchDirector}[2]{\HECdirecteurRecherche{#1}{#2}}

\newcommand{\HECcodirecteurRecherche}[2]{%
 \renewcommand{\HEC@codirecteurRecherche}{#1}
 \ifthenelse{\equal{#2}{F}}{%
  \setboolean{HEC@isCodirRechFemme}{true}
 }{%
  \setboolean{HEC@isCodirRechFemme}{false}
 }
}
\newcommand{\HECresearchCodirector}[2]{\HECcodirecteurRecherche{#1}{#2}}

\newcommand{\HECuniversiteCodirecteur}[1]{%
 \renewcommand{\HEC@universiteCodirecteur}{#1}
}
\newcommand{\HECcodirectorUniversity}[1]{\HECuniversiteCodirecteur{#1}}

\newcommand{\HECmembreJury}[1]{%
 \renewcommand{\HEC@membreJury}{#1}
}
\newcommand{\HECjuryMember}[1]{\HECmembreJury{#1}}

\newcommand{\HECuniversiteMembreJury}[1]{%
 \renewcommand{\HEC@universiteMembreJury}{#1}
}
\newcommand{\HECjuryMemberUniversity}[1]{\HECuniversiteMembreJury{#1}}

\newcommand{\HECexaminateurExterne}[2]{%
 \renewcommand{\HEC@examinateurExterne}{#1}
 \ifthenelse{\equal{#2}{F}}{%
  \setboolean{HEC@isExamExtFemme}{true}
 }{%
  \setboolean{HEC@isExamExtFemme}{false}
 }
}
\newcommand{\HECexternalExaminator}[2]{\HECexaminateurExterne{#1}{#2}}

\newcommand{\HECuniversiteExaminateur}[1]{%
 \renewcommand{\HEC@universiteExaminateur}{#1}
}
```

```
\newcommand{\HECexaminatorUniversity}[1]{\HECuniversiteExaminateur{#1}}

\newcommand{\HECrepresentantDirecteur}[2]{%
 \renewcommand{\HEC@representantDirecteur}{#1}
 \ifthenelse{\equal{#2}{F}}{%
  \setboolean{HEC@isRepDirFemme}{true}
 }{%
  \setboolean{HEC@isRepDirFemme}{false}
 }
}
\newcommand{\HECdirectorRepresentative}[2]{\HECrepresentantDirecteur{#1}{#2}}

```

A.6.2 pdf metadata

Along with the metadata-related commands for your work, the class defines commands that will insert metadata in the `.pdf` file generated after the compilation of your dissertation or thesis. These commands are placed in the `hyperref` package's options.

```

\newcommand{\HECpdfauteur}{\HEC@auteur}
\newcommand{\HECpdftitre}{\HEC@titre}

```

A.6.3 Title and jury identification pages

The class uses three internal commands to generate the title page and the jury identification page. The `\HECtitlepages` command is inserted at the beginning of the `document` environment to generate the title pages related to each type of document written.

`\HEC@pageTitrePhD` The `\HEC@pageTitrePhD` command generates the dissertations' title page. It uses the `memoir` class' `titlingpage` environment, which gives more flexibility than LaTeX's `\maketitle` command for the creation of title pages[5]. The `titlingpage` environment resets the page numbering at 1 after the title page so it prevents the following blank page from being counted as a page.

Automatic insertion of the subtitle is made possible by calculating its length. If it is empty, only a new line is inserted. If it's not empty, a colon is inserted followed by the subtitle.

Instead of defining fixed vertical spaces between the title page elements, the command makes an extensive use of the `\vfill` command, which vertically justifies all elements in the page, no matter what size they are.

```

\newcommand{\HEC@pageTitrePhD}{%
 \begin{titlingpage}
  \centering
  \begin{SingleSpace}
   {\Large HEC MONTR\'{E}AL}\\
```

```
  \'{E}cole affili\'{e}e \`{a} l'Universit\'{e} de Montr\'{e}al
  \vfill
  {\bfseries\HEC@titre
   \ifthenelse{\equal{\HEC@sousTitre}{}}%
    { \\ }%
    {~: \\ \HEC@sousTitre}
   \vfill
   par \\
   \HEC@auteur}
  \vfill
  Th\`{e}se pr\'{e}sent\'{e}e en vue de l'obtention du grade de Ph. D. en administration \\
  (sp\'{e}cialisation \HEC@optionPhD)
  \vfill
  \HEC@moisDepot~\HEC@anneeDepot
  \vfill
  \copyright~\HEC@auteur, \HEC@anneeDepot
 \end{SingleSpace}
\end{titlingpage}
}
```

`\HEC@pageIdentificationJury` This command uses the `titlingpage` environment's starred version which doesn't reset the page numbering at 1. In this way, the french abstract can start at page number iii, as requested.

The command also determines each contributor's job title spelling according to his or her gender. It is done by evaluating each `HEC@is*Femme` boolean variable's value.

```

\newcommand{\HEC@pageIdentificationJury}{%
 \begin{titlingpage*}
  \centering
  \begin{SingleSpace}
   {\Large HEC MONTR\'{E}AL}\\
   \'{E}cole affili\'{e}e \`{a} l'Universit\'{e} de Montr\'{e}al
   \vfill
   Cette th\`{e}se intitul\'{e}e :
   \vfill
   {\bfseries\HEC@titre
    \ifthenelse{\equal{\HEC@sousTitre}{}}%
    { \\ }%
    {~: \\ \HEC@sousTitre}}
   \vfill
   Pr\'{e}sent\'{e}e par :
   \vfill %
   {\bfseries \HEC@auteur}
   \vfill
   a \'{e}t\'{e} \'{e}valu\'{e}e par un jury compos\'{e} des personnes suivantes :
   \vfill
   \HEC@presidentRapporteur \\
   HEC Montr\'{e}al \\
   \ifthenelse{\boolean{HEC@isPresRappFemme}}%
    {Pr\'{e}sidente-rapportrice}%
```

```
    {Pr\'{e}sident-rapporteur}
   \vfill
   \HEC@directeurRecherche \\
   HEC Montr\'{e}al \\
   \ifthenelse{\boolean{HEC@isDirRechFemme}}%
    {Directrice de recherche}%
    {Directeur de recherche}
   \vfill
   \HEC@codirecteurRecherche \\
   \HEC@universiteCodirecteur \\
   \ifthenelse{\boolean{HEC@isCodirRechFemme}}%
    {Codirectrice de recherche}%
    {Codirecteur de recherche}
   \vfill
   \HEC@membreJury \\
   \HEC@universiteMembreJury \\
   Membre du jury
   \vfill
   \HEC@examinateurExterne \\
   \HEC@universiteExaminateur \\
   \ifthenelse{\boolean{HEC@isExamExtFemme}}%
    {Examinatrice externe}%
    {Examinateur externe}
   \vfill
   \HEC@representantDirecteur \\
   HEC Montr\'{e}al \\
   \ifthenelse{\boolean{HEC@isRepDirFemme}}{%
    Repr\'{e}sentante du directeur de HEC Montr\'{e}al}{%
    Repr\'{e}sentant du directeur de HEC Montr\'{e}al}
  \end{SingleSpace}
 \end{titlingpage*}
}

```

`\HEC@pageTitreMSc` The `\HEC@pageTitreMSc` command uses the `titlingpage` environment in the same manner as the dissertations' title page command.

This command is also the only place where a fixed vertical space is inserted between the title and the subtitle, in order to comply with the *Guidelines*' presentation standards.

```

\newcommand{\HEC@pageTitreMSc}{%
 \begin{titlingpage}
  \centering
  \begin{SingleSpace}
   {\Large HEC MONTR\'{E}AL}
   \vfill
   {\bfseries\HEC@titre
    \ifthenelse{\equal{\HEC@sousTitre}{}}%
     {\\[12pt]}%
     {~: \\ \HEC@sousTitre \\[12pt]}
    par \\[12pt]
    \HEC@auteur
   \vfill
```

```
    \HEC@directeurRecherche \\
    HEC Montr\'{e}al \\
    \ifthenelse{\boolean{HEC@isDirRechFemme}}%
    {Directrice de recherche}%
    {Directeur de recherche}
     \vfill %
     Sciences de la gestion \\%
     (Sp\'{e}cialisation \HEC@optionMSc)}
    \vfill
    \emph{M\'{e}moire pr\'{e}sent\'{e} en vue de l'obtention \\ %
     du grade de ma\^{i}trise \`{e}s sciences \\ %
     (M. Sc.)}
    \vfill
    \HEC@moisDepot~\HEC@anneeDepot \\ %
    \copyright~\HEC@auteur, \HEC@anneeDepot
   \end{SingleSpace}
  \end{titlingpage}
}
```

`\HECtitlepages` The command evaluates the `HEC@isPhD` boolean variable's value. If the document is a dissertation, it inserts the dissertation's title page and the jury identification page. If the document is a thesis, the command inserts the thesis' title page.

```
\newcommand{\HECpagestitre}{%
 \ifthenelse{\boolean{HEC@isPhD}}{%
  \HEC@pageTitrePhD
  \HEC@pageIdentificationJury
 }{%
  \HEC@pageTitreMSc
 }
}
\newcommand{\HECtitlepages}{\HECpagestitre}
```

A.6.4 Introduction and conclusion titles

In dissertations and theses written with articles, there is more than one introduction and conclusion : a general introduction and conclusion for the whole document, and one introduction and conclusion for every article. To make a distinction between all of these introductions and conclusions, the class changes the document's introduction's and conclusion's titles to "General Introduction" and "General Conclusion". The command also takes into account the document's default language, as for all other titles, as we'll see it in Section A.6.5.

```
\newcommand{\HECtitreIntroduction}{Introduction}
\newcommand{\HECtitreConclusion}{Conclusion}
\newcommand{\HECgenererTitres}{%
 \ifthenelse{\boolean{HEC@isClassique}}{}{%
  \IfLanguageName{english}{%
   \renewcommand{\HECtitreIntroduction}{General Introduction}
```

```
   \renewcommand{\HECtitreConclusion}{General Conclusion}
  }{%
   \renewcommand{\HECtitreIntroduction}{Introduction g\'{e}n\'{e}rale}
   \renewcommand{\HECtitreConclusion}{Conclusion g\'{e}n\'{e}rale}
  }
 }
}

```

A.6.5 Automatic english translation of titles and table of contents

The `iflang` package[13] enables the detection of a document's default language and allows the class to carry out conditional actions depending on the language. The **hecthese** class supports French and English and uses `iflang` to generate "homemade" section titles for dissertations and theses.

```

\newcommand{\HECtdmAbreviations}{%
 \IfLanguageName{english}{List of Acronyms}{Liste des abr\'{e}viations}
}

\newcommand{\HECtdmRemerciements}{%
 \IfLanguageName{english}{Acknowledgements}{Remerciements}
}

\newcommand{\HECtdmAvantPropos}{%
 \IfLanguageName{english}{Preface}{Avant-propos}
}

\newcommand{\HECtdmCadreTheorique}{%
 \IfLanguageName{english}{Theoretical Framework}{Cadre th\'{e}orique}
}

\newcommand{\HECtdmRevueLitterature}{%
 \IfLanguageName{english}{Literature Review}{Revue de la litt\'{e}rature}
}

\newcommand{\HECtdmResumeArticle}{%
 \IfLanguageName{english}{Abstract}{R\'{e}sum\'{e}}
}

```

A.7 Class environments

The `HECdedication` environment creates a text block that is vertically centered, right-aligned horizontally and that takes only half of the page's regular text width. The text block is also emphasized.

The `HECabbreviations` environment is a variation of the `description` environment. It creates an acronyms list on two aligned columns : one for the acronyms and one for their definition.

```

```

```
\newenvironment{HECdedicace}{%
 \vfill
 \hfill
 \begin{minipage}{0.5\textwidth}
  \itshape}%
 {%
 \end{minipage}
 \vfill%
}
\newenvironment{HECdedication}{\begin{HECdedicace}}{\end{HECdedicace}}

\newenvironment{HECabreviations}[1]{%
 \begin{description}[leftmargin=!,labelwidth=\widthof{\bfseries #1}]}%
 {%
 \end{description}%
}
\newenvironment{HECabbreviations}[1]{%
 \begin{HECabreviations}{#1}}%
 {\end{HECabreviations}}

```

A.8 Package options

The official French translations for the list of figures and the index don't correspond to those of the *Guidelines*' French version. These translations are overridden by the class so they match with the university's official labels.

```

\addto\captionsfrench{%
 \renewcommand{\listfigurename}{Liste des figures}
 \renewcommand{\indexname}{Index analytique}
}
⟨/class⟩
```

References

[1] Centre de formation en langues des affaires (2020). *Guide pour la rédaction d'un travail universitaire de 1er, 2e et 3e cycles*, HEC Montréal. Consulté le 24 août 2020 à https://www.hec.ca/cfla/francais/document/guide-redaction-travail-cycles.pdf

[2] Centre de formation en langues des affaires (2014). *Guidelines for writing and academic work at a graduate level*, HEC Montréal. Consulté le 24 avril 2018 à https://www.hec.ca/cfla/anglais/document/guidelines_academic_work.pdf

[3] *TeX Live*, TeX Users Group. Consulté le 18 mai 2017 à https://www.tug.org/texlive/

[4] van der Zander, Benito, Jan Sundermeyer, Daniel Braun et Tim Hoffmann (2017). *TeXstudio : LaTeX made comfortable*, TeXstudio. Consulté le 18 mai 2017 à http://www.texstudio.org/

[5] Wilson, Peter R., Lars Madsen (2016). *Package memoir*, Comprehensive TeX Archive Network. Consulté le 19 mai 2017 à https://www.ctan.org/pkg/memoir

[6] Bezos López, Javier, Johannes L. Braams (2017). *Package babel*, Comprehensive TeX Archive Network. Consulté le 19 mai 2017 à https://www.ctan.org/pkg/babel

[7] Oberdiek, Heiko, Sebastian Rahtz (2017). *Package hyperref*, Comprehensive TeX Archive Network. Consulté le 19 mai 2017 à https://www.ctan.org/pkg/hyperref

[8] Thorup, Kresten Krab, Frank Jensen et The LaTeX Team (2007). *Package calc*, Comprehensive TeX Archive Network. Consulté le 23 mai 2017 à https://www.ctan.org/pkg/calc

[9] Bezos López, Javier (2009). *Package enumitem*, Comprehensive TeX Archive Network. Consulté le 23 mai 2017 à https://www.ctan.org/pkg/enumitem

[10] Lamport, Leslie, David Carlisle et The LaTeX Team (2014). *Package ifthen*, Comprehensive TeX Archive Network. Consulté le 23 mai 2017 à https://www.ctan.org/pkg/ifthen

[11] Goulet, Vincent (2012). *Package francais-bst*, Comprehensive TeX Archive Network. Consulté le 23 mai 2017 à https://www.ctan.org/pkg/francais-bst

[12] Daly, Patrick W., Arthur Ogawa (2009). *Package natbib*, Comprehensive TeX Archive Network. Consulté le 29 mai 2017 à https://www.ctan.org/pkg/natbib

[13] Oberdiek, Heiko (2007). *Package iflang*, Comprehensive TeX Archive Network. Consulté le 25 septembre 2017 à https://www.ctan.org/pkg/iflang

[14] Archambault, Caroline (2017). *Bibliographie selon le style HEC Montréal*, Bibliothèque HEC Montréal. Consulté le 5 octobre 2017 à http://libguides.hec.ca/style-hec

[15] Secrétariat général (2018). *Réglement pédagogique de la faculté des études supérieures et postdoctorales*, Université de Montréal. Consulté le 23 avril 2018 à http://secretariatgeneral.umontreal.ca/documents-officiels/reglements-et-politiques/reglement-pedagogique-de-la-faculte-des-etudes-superieures-et-postdoctorales/

[16] Faculté des études supérieures et postdoctorales (2015). *Guide de présentation des mémoires et des thèses de l'Université de Montréal*, Université de Montréal. Consulté le 23 avril 2018 à http://www.fesp.umontreal.ca/fileadmin/fesp/documents/Cheminement/GuidePresentationMemoiresTheses.pdf

[17] Kime, Philip, Moritz Wemheuer (2023). *Package biblatex*, Comprehensive TeX Archive Network. Consulté le 17 octobre 2023 à https://www.ctan.org/pkg/biblatex

Change History

0.2
General: Première version bêta lancée pour les tests usagers . . 1

0.3
General: Retrait du package `mathptmx` des packages requis par la classe. Ajout du choix de polices entre `mathptmx` et `mathpazo` dans les gabarits. Ajout des sections «Rédaction» et «Compilation» dans la documentation. 1

0.4
General: Retrait de la section «Mise en garde» de la documentation et ajout de l'encadré «Phase de tests de la version bêta. Lancement du package pour fins de tests. 1

0.4.1
General: Ajout d'une condition pour vérifier la présence du sous-titre dans la mise en forme de la page d'identification du jury. 1

1.0
General: Première version de production lancée publiquement. Ajout du préfixe HEC à toutes les commandes publiques. Révision, correction et augmentation de la documentation. Ajout des sections non numérotées "mots clés" et "méthodes de recherche" dans le résumé et l'abstract. Prise en charge de l'anglais pour les intitulés de chapitres et sections. Correction du bogue d'affichage des citations "hors articles" dans les thèses et mémoires par articles. 1

1.1
General: Uniformisation de la licence lppl dans tous les fichiers. Ajout de liens vers des capsules vidéos dans la documentation. 1

1.2
General: Mise à jour des liens vers les capsules vidéos dans la documentation. 1

1.3
General: Première version bilingue de la classe / First bilingual version of the class 1

1.3.1
General: Ajout d'une note pour les utilisateurs de MacOS en ce qui concerne l'installation de la classe. / Added a note to MacOS users concerning the class installation. 1

1.3.2
General: Coquille dans la dtx et changement du lien vers le gabarit anglais pour MacOS. . . 1

1.4
General: Beaucoup de corrections de bogues et de mises à jour (voir dépôt Git). / Lots of bug fixes and updates (see Git repository). 1

1.5
General: Corrigé l'implémentation de chapterbib. / Fixed the chapterbib implementation. . . . 1

1.5.1
General: Corrigé la typographie de certaines phantomsections / Corrected typos in phantomsections 1

2.0
General: Ajout de biblatex et abandon de chapterbib / Added biblatex and dropped chapterbib 1

2.1
General: Ajout de la section Contribution de l'étudiant / Added Student Contribution Section 1